

ASSIGNMENT #07

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Tuesday, May 5th in Class

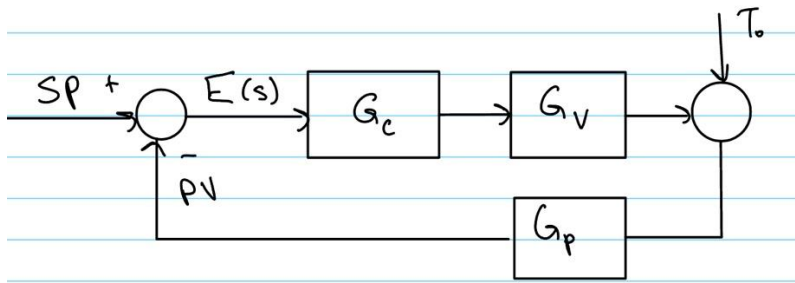
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name in the report.
 - Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
 - Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
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Problem 1: Feedback Control of a Thermal System

1. The block diagram looks as follows,



Where the G_v is approx. 1, $G_c = 1500(0.5s + 1)$, $G_p = \frac{1}{(\tau s + 1)}$. G_p encompasses all three of the tanks.

2. $\frac{PV}{SP} = \frac{G_c G_p}{1 + G_c G_p}$ where the G_v term is neglected as it is equal to one.
3. $\frac{PV}{T_0} = \frac{G_p}{1 + G_c G_p}$
4. T_0 in the Laplace space would be equal to $1/s$, making the equation from answer 3 now becoming,

$$PV = \frac{G_p}{1 + G_c G_p} * \frac{1}{s}$$

Taking the limit of sPV as s goes to 0 would then become,

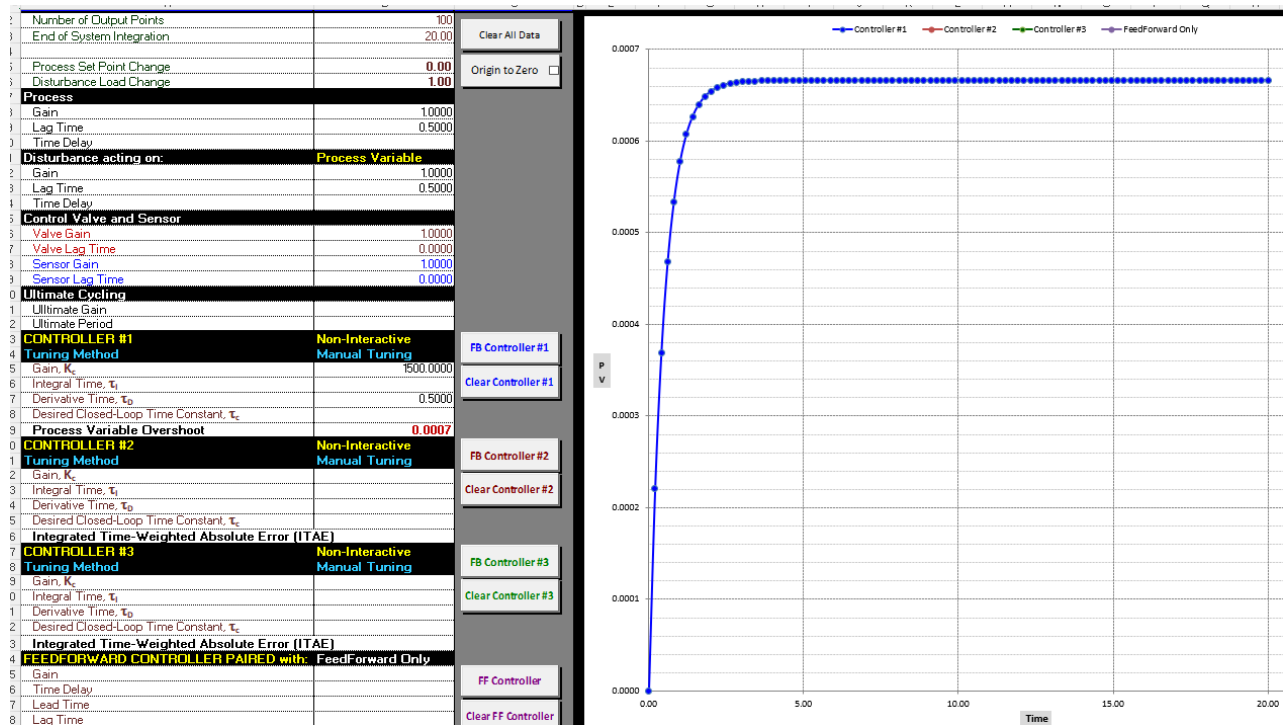
$$\lim_{s \rightarrow 0} PV = \frac{G_p}{1 + G_c G_p}$$

Which simplifies to,

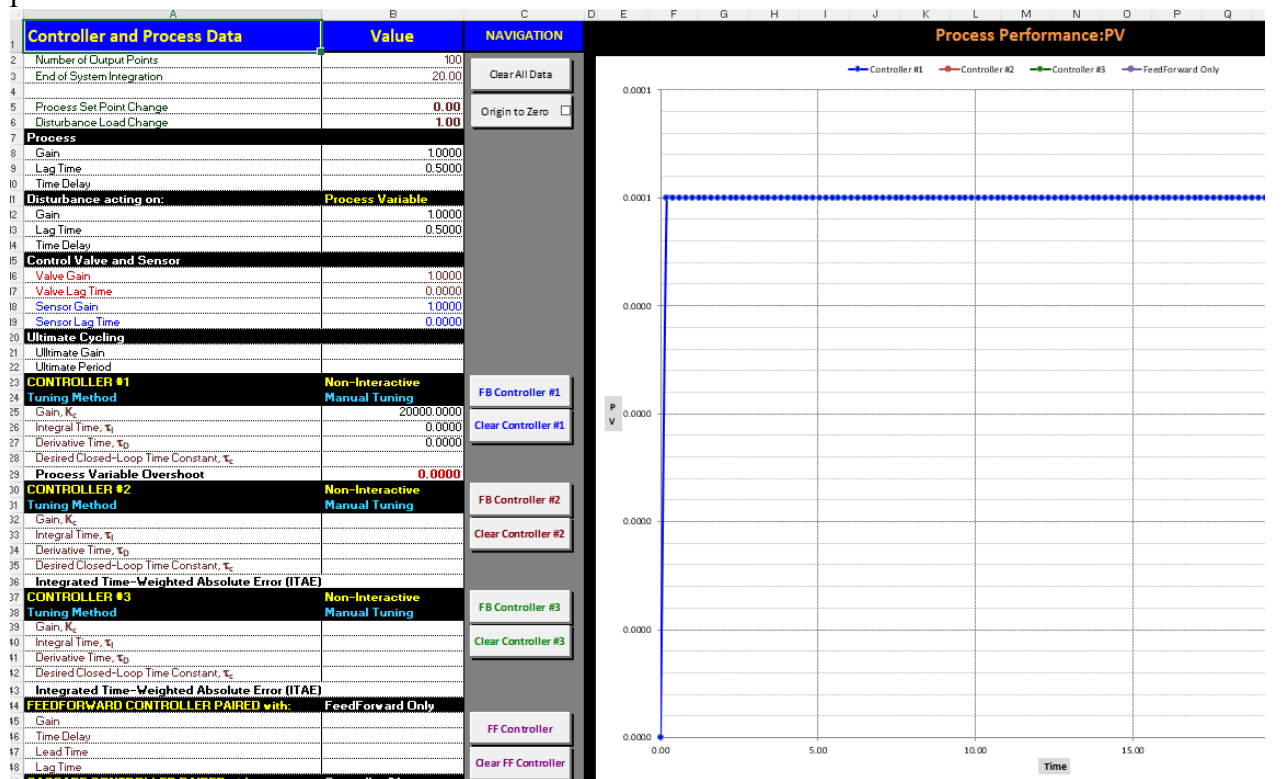
$$\lim_{s \rightarrow 0} PV = \frac{1}{(\tau s + 1) + 1500(0.5s + 1)}$$

With an answer of 6.6×10^{-4}

5. When using the **CHE Calculator** Automation panel, an overshoot of 7×10^{-4} was gotten, reference the screenshot below. Note that the process/disturbance lag time was approximated at 0.5 to mimic the controller function, and a gain of one was used. This aligns well with the answer found in part four.



6. Technically, if the controller gain was increased to 20000 and the derivative time was removed, the calculated offset would be zero – **however**, it is unlikely that this controller would be very stable. I would recommend using the previous controller settings for the process.



7. The following table summarizes results provided in answers 5 and 6.

	Controller for #5	Controller for #6
K_C	1500	20000
Tau_D	0.5	0
Settling Time	3.2323	0.2020

Problem 2: Feedforward Control of a Distillation Column Tray Temperature

- a. Using the provided model, $G_{FF} = -\frac{G_d}{G_p} = 4e^{-13s} \frac{6s+1}{30s+1}$

Which should be realizable, as it has a feasible time delay.

Using the equation provided within part (a), the following simplifications must be done.

$$\frac{T}{F} = \frac{-4e^{-2s} L}{6s+1} \frac{1}{F} + \frac{25e^{-15s}}{30s+1}$$

$$\frac{T}{L} = \frac{-4e^{-2s}}{6s+1} + \frac{25e^{-15s} F}{30s+1} \frac{1}{L}$$

Then, becoming T/F over T/L, the simplified answer becomes,

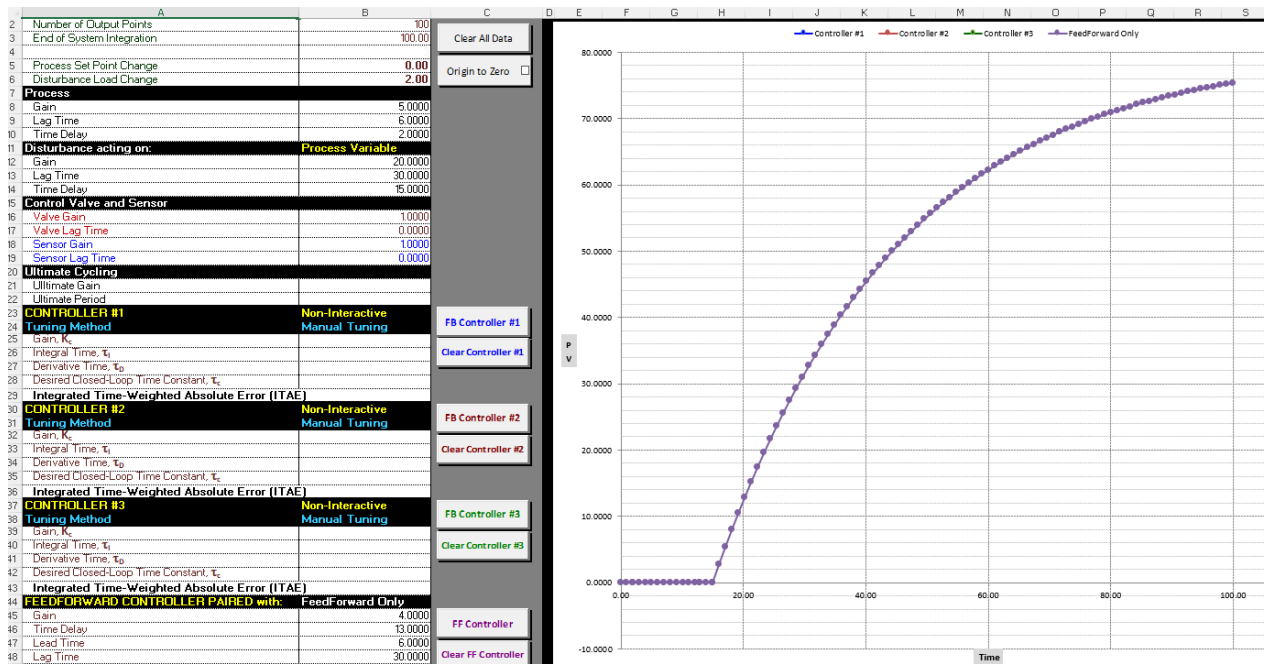
$$\frac{T/F}{T/L} = \frac{1}{1 - \frac{25e^{-15s}(6s+1)}{4e^{-2s}(30s+1)}} \left(\frac{L}{F}\right)^2 + \frac{1}{\frac{-4e^{-2s}(30s+1)}{25e^{-15s}(6s+1)} + 1} \left(\frac{L}{F}\right)$$

Or

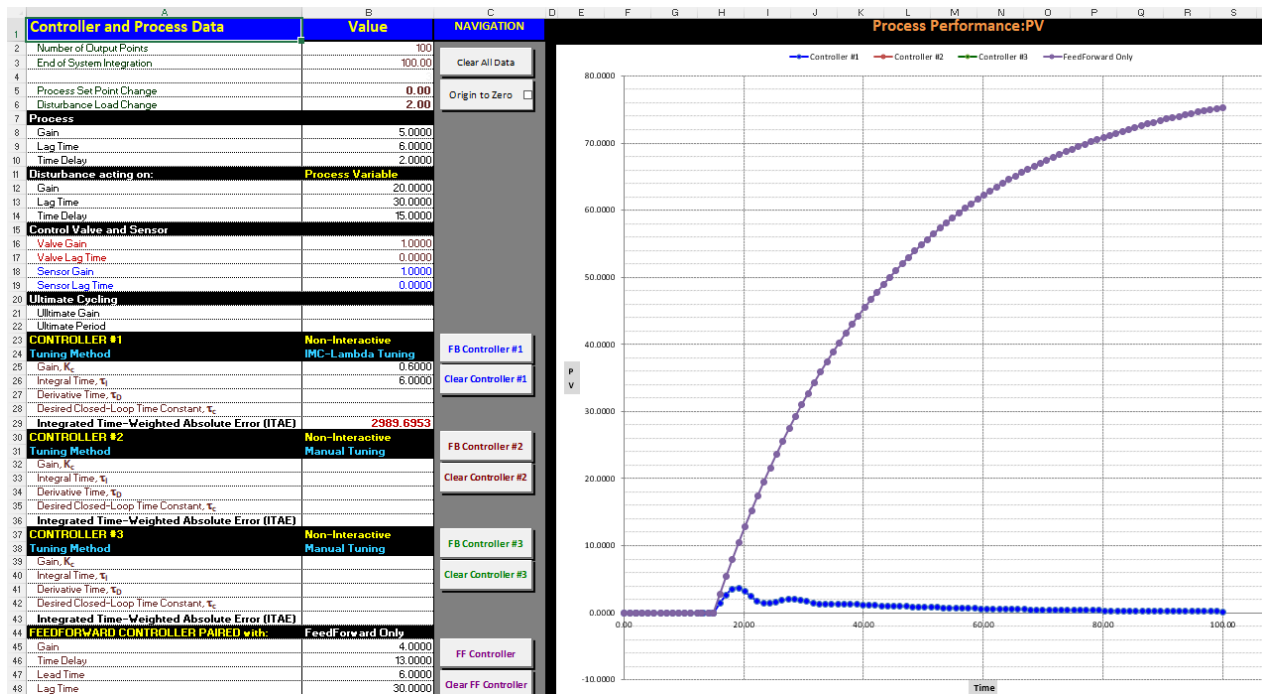
$$G_{FF} = \frac{1}{1 - \frac{25e^{-13s}(6s+1)}{4(30s+1)}} \left(\frac{L}{F}\right)^2 + \frac{1}{\frac{-4e^{13s}(30s+1)}{25(6s+1)} + 1} \left(\frac{L}{F}\right)$$

Which is not in a realizable format.

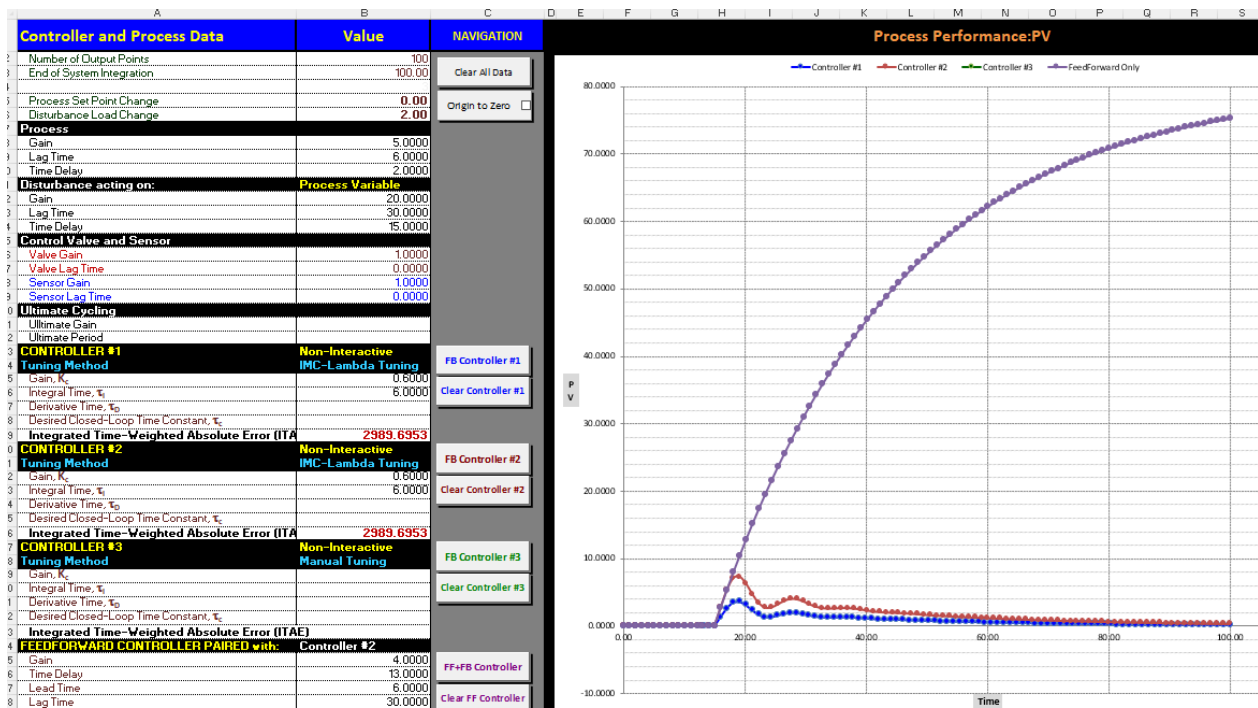
b.



c.



d & e.



Using both the feed forward and feed back controllers resulted in the process having a lower overshoot and a consistently low overshoot value until the system settles.

Normally with a feed forward and a feed back controller there would be virtually no disruption within the process, however that is not the case for this process.